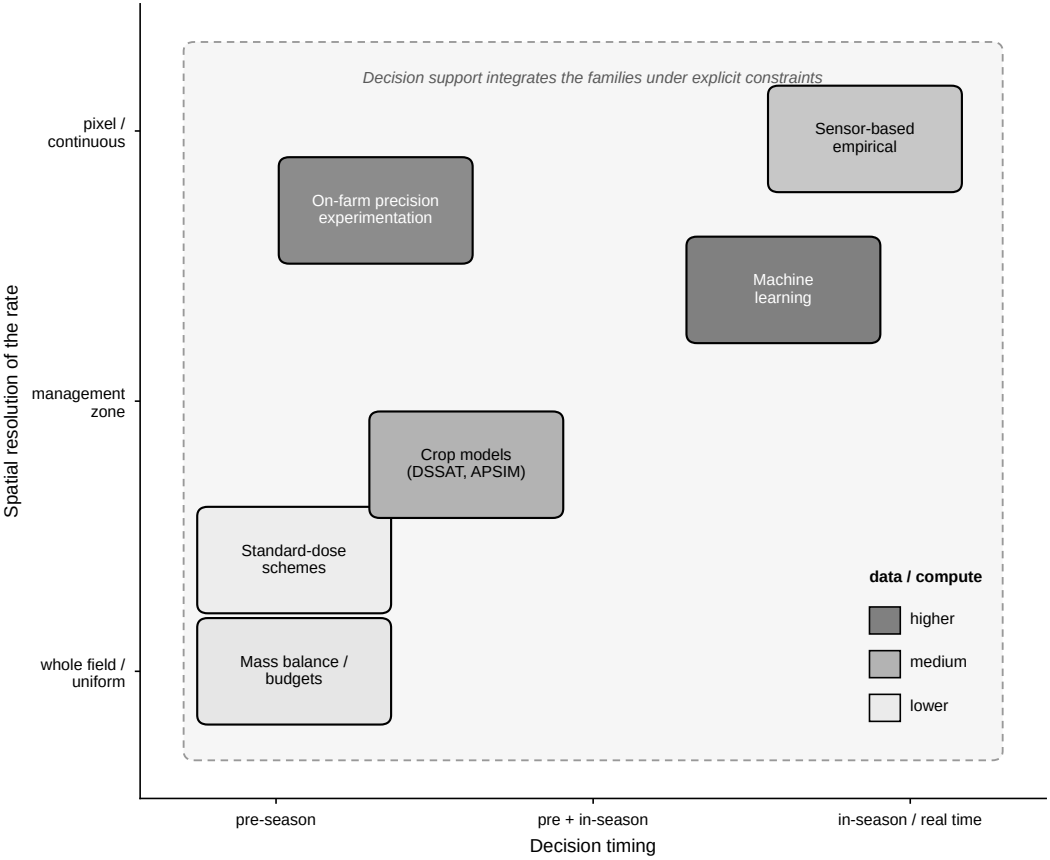
Decision support integrates the families under explicit constraints
Sensor-based empirical
On-farm precision experimentation
Machine learning
Crop models (DSSAT, APSIM)
Standard-dose schemes
Mass balance / budgets
data / compute
higher
medium
lower
pixel / continuous
management zone
whole field / uniform
Spatial resolution of the rate
pre-season
pre + in-season
in-season / real time
Decision timing